Descriptive Statistics:-

1. What is Descriptive Statistics?

Descriptive Statistics is a branch of statistics that is used to **collect, organize, summarize, analyze, and present data** in a meaningful way.

It helps us understand the main characteristics of a dataset without drawing conclusions about the entire population.

Simple Definition

Descriptive Statistics is the process of summarizing data using numbers, tables, and graphs to understand its main characteristics.

Why is Descriptive Statistics Important?

Imagine a company has data of **1,00,000 customers**.

Without descriptive statistics, it is almost impossible to understand the data.

Using descriptive statistics, we can easily answer questions like:

- What is the average customer age?
- What is the highest salary?
- What is the lowest mark?
- How much variation exists in sales?
- Which product is sold the most?

Why is it Used in Machine Learning?

Before building any Machine Learning model, we first perform **Exploratory Data Analysis (EDA)**.

Descriptive statistics helps us:

- Understand the dataset
- Detect missing values
- Detect outliers
- Understand feature distribution
- Check data quality
- Select useful features

- Decide preprocessing techniques
- Improve model accuracy

It is usually the **first step in every Data Science project**.

Types of Descriptive Statistics

Descriptive statistics is divided into three major categories:

1. Measures of Central Tendency
2. Measures of Dispersion (Spread)
3. Measures of Shape

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1. Measures of Central Tendency

These measures describe the **center** of the dataset.

They answer:

"What is the typical value?"

There are three measures:

- Mean
- Median
- Mode

Mean (Average)

Definition

The mean is the sum of all observations divided by the number of observations.

Original amounts

Equal shares

2

4

7

$\bar{x}$ = 4 1/3

Average of 2, 4, and 7 is (2 + 4 + 7) / 3 = 13 / 3 = 4 1/3

$$x_1$$

$$x_2$$

$$x_3$$

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

where:

- ΣX = Sum of observations
- N = Number of observations

Example

Marks

50 60 70 80 90

Step 1

Sum = 50+60+70+80+90

=350

Step 2

Mean=350/5

=70

Python

```
import numpy as np

marks=[50,60,70,80,90]

print(np.mean(marks))
```

Output

70

Advantages

- Easy to calculate
- Uses all observations
- Most commonly used

Disadvantages

- Affected by outliers
- Not suitable for skewed data

Median

Definition

Median is the **middle value** after arranging data in ascending order.

Mean38

Median38

0

25

50

75

100

No outlier: mean is 38; median is 38

No outlierHigh outlier

No outlierHigh outlier

Example 1

10 20 30 40 50

Middle value

30

Median = 30

Example 2

10 20 30 40

Median

(20+30)/2

=25

Python

```
import numpy as np

marks=[10,20,30,40]

print(np.median(marks))
```

Output

25

Advantages

- Not affected by outliers
- Best for skewed data

Disadvantages

- Ignores many values when determining the center
- Less useful for mathematical calculations than the mean

Mode

Definition

Mode is the value that appears **most frequently**.

Example

2 2 3 4 5 5 5 6

Mode

5

Python

```
from scipy import stats

marks=[2,2,3,4,5,5,5,6]

print(stats.mode(marks))
```

Advantages

- Useful for categorical data

- Easy to understand

Disadvantages

- May have multiple modes
- Sometimes no mode exists

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2. Measures of Dispersion

These measure how spread out the data is.

Example

Student A

70 70 70 70 70

Student B

40 55 70 85 100

Both have the same mean, but Student B's marks vary much more.

Measures of dispersion quantify this variation.

Types

- Range
- Variance
- Standard Deviation
- Interquartile Range (IQR)

Range

Formula

Range = Maximum − Minimum

Example

20 30 40 60 80

Range

$=80-20$

=60

Python

```
marks=[20,30,40,60,80]

print(max(marks)-min(marks))
```

Variance

Definition

Variance measures the average squared distance of each observation from the mean.

$$Var(X) = E(X^2) - [E(X)]^2$$

$$Var(X) = (1.4)^2 = 1.96$$

$$\sigma$$

μ-σ+σVar(X) ≈ 1.96

Example

Marks

60 65 70 75 80

A small variance means the values are close to the mean. A large variance means they are more spread out.

Python

```
import numpy as np

marks=[60,65,70,75,80]

print(np.var(marks))
```

Standard Deviation

Definition

Standard deviation is the square root of variance. It tells us how much the values typically differ from the mean.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

The points have mean μ = 5 and σ = 1.5.

$$\sigma$$

Drag points

Interpretation

Low Standard Deviation

Values are close together.

High Standard Deviation

Values are widely spread.

Python

```
import numpy as np

marks=[60,65,70,75,80]

print(np.std(marks))
```

Why is Standard Deviation Important?

It is used in:

- Feature Scaling
- Outlier Detection
- Z-score Calculation
- Machine Learning Algorithms
- Risk Analysis

Interquartile Range (IQR)

Formula

IQR = Q3 − Q1

Where:

- Q1 = First Quartile (25%)
- Q3 = Third Quartile (75%)

Uses

- Detecting Outliers

- Robust measure of spread
- Boxplots

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3. Measures of Shape

These describe the overall shape of the data distribution.

Skewness

Definition

Skewness measures whether the distribution is symmetric or tilted to one side.

-6-4-22460.10.20.30.4ValueDensityMeanMedianMode

Right skew: mode, median, mean from peak to tail

Direction

LeftRight

LeftRight

Skew strength

Types

Symmetric Distribution

Mean = Median = Mode

Positive (Right) Skew

- Long tail on the right
- Mean > Median

Examples:

- Salaries
- House prices

Negative (Left) Skew

- Long tail on the left
- Mean < Median

Examples:

- Easy exam scores

Python

print(df.skew())

Kurtosis

Definition

Kurtosis measures the heaviness of the tails or the presence of extreme values.

Types

- Mesokurtic (Normal)
- Leptokurtic (Heavy tails)
- Platykurtic (Light tails)

Python

print(df.kurt())

Five Number Summary

A quick summary of a dataset consists of:

Statistic	Meaning
Minimum	Smallest value
Q1	25th percentile
Median	50th percentile
Q3	75th percentile
Maximum	Largest value

Descriptive Statistics in Pandas

```
import pandas as pd

data={
   "Marks":[45,55,67,78,89,91,95]
}

df=pd.DataFrame(data)

print(df.describe())
```

Output includes:

- Count

- Mean
- Standard Deviation
- Minimum
- 25%
- 50%
- 75%
- Maximum

Useful Python Functions

Function	Description
mean()	Average
median()	Middle value
mode()	Most frequent value
max()	Largest value
min()	Smallest value
std()	Standard deviation
var()	Variance
describe()	Summary statistics
quantile()	Quartiles
skew()	Skewness
kurt()	Kurtosis

Real-Life Example

Suppose a school has the marks of 500 students.

Using descriptive statistics, the school can determine:

- Average marks
- Highest and lowest marks
- Most common marks
- Students with unusually low or high scores

- Whether marks are evenly distributed or skewed
- How much variation exists in performance

This information helps teachers evaluate overall class performance before making academic decisions.

Advantages

- Summarizes large datasets quickly
- Easy to understand
- Supports visualization and reporting
- Detects anomalies such as outliers
- Forms the foundation of Exploratory Data Analysis (EDA)
- Essential before applying Machine Learning algorithms

Limitations

- Describes only the available data
- Does not explain cause-and-effect relationships
- Cannot make predictions about future or unseen data
- Does not infer characteristics of the entire population